

Task for BD07
Farré Vicente Josep
Bases de dades

Task for BD07

LINK GIFTHUB:

LINK LOOM: <https://www.loom.com/share/43ee4411037c455eb0c8b60cd4b33329>

(Gaire bé he pogut explicar els 16 exercicis, he tingut que repetir-ho 6/7 cops per què no me donava temps en 5 min i amb el meu anglès, me sap greu)

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1. Create a database called zoo.

Answer: In MongoDB, databases are created automatically when selected with the [use](#) command.

```
test> use zoo  
switched to db zoo  
zoo> db.createCollection("animals")  
{ ok: 1 }
```

2. Insert a collection called animals.

A: Collections in MongoDB can be created explicitly with [CreateCollection](#) ("")

```
switched to db zoo  
zoo> db.createCollection("animals")  
{ ok: 1 }
```

3. Confirm that the database and the collection have been successfully created.

A: To verify the active database and existing collections

[db](#) → Shows the active database

[show dbs](#) → Lists all databases

[show collections](#) → Lists collections in the active database

```
zoo> db  
zoo  
zoo> show dbs  
admin    40.00 KiB  
config   72.00 KiB  
local    40.00 KiB  
zoo       8.00 KiB  
zoo> show collections  
animals  
zoo> |
```

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4. Insert the documents contained in the animals.json file

A: There are 2 options:

Option A: Using `insertMany` inside MongoDBshell

```
db.animals.insertMany([
  { _id: 1, name: "Shere Khan", age: 10, gender: "Male", species: "Tiger",
    habitat: "Rainforest", birthplace: "Sundarbans, India",
    parents: [{ name: "Unknown", gender: "Male" }, { name: "Unknown", gender:
    "Female" }],
    caretakers: [{ idCaretaker: 101, nameCaretaker: "John Smith" }] },
  { _id: 2, name: "Rajah", age: 8, gender: "Male", species: "Tiger", habitat:
    "Rainforest", birthplace: "Bandipur Tiger Reserve, India", parents: [{ name:
    "Shere Khan", gender: "Male" }, { name: "Tala", gender: "Female" }],
    caretakers": [{ idCaretaker: 102, nameCaretaker: "Emily Brown" }] },
```

.....

And the rest of ID's

Option B: Importing with mongoimport

```
mongoimport --db zoo --collection collection animals --file "rute"/animals.json
--jsonArray
```

or in MongoDBcompass import your json.

5- Write a MongoDB query to find animals that have Rainforest as their habitat and are attended by a caretaker named Emily Brown.

A: Query filtering by habitat 'Rainforest' and caretaker 'Emily Brown' using the `$elemMatch` operator:

```
db.animals.find({ habitat: "Rainforest", caretakers: { $elemMatch: {
nameCaretaker: "Emily Brown" } } })
```

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```
zoo> db.animals.find({ habitat: "Rainforest", caretakers: { $elemMatch: { nameCaretaker: "Emily Brown" } } })
[
  {
    _id: 2,
    name: 'Rajah',
    age: 8,
    gender: 'Male',
    species: 'Tiger',
    habitat: 'Rainforest',
    birthplace: 'Bandipur Tiger Reserve, India',
    parents: [
      { name: 'Shere Khan', gender: 'Male' },
      { name: 'Tala', gender: 'Female' }
    ],
    caretakers: [ { idCaretaker: 102, nameCaretaker: 'Emily Brown' } ]
  }
]
```

6- Write a MongoDB query to display only the name, species, and habitat of all animals sorted by their species in ascending order.

A: Projection with the requested fields (1 = show, 0 = hide) and ascending sort by **species**:

```
db.animals.find( {}, { _id: 0, name: 1, species: 1, habitat: 1 } ).sort({
species: 1 })
```

```
zoo> db.animals.find( {}, { _id: 0, name: 1, species: 1, habitat: 1 } ).sort({ species: 1 })
[
  { name: 'Balu', species: 'Bear', habitat: 'Mountain Forest' },
  { name: 'Koda', species: 'Bear', habitat: 'Mountain Forest' },
  { name: 'Kenai', species: 'Bear', habitat: 'Mountain Forest' },
  { name: 'Smokey', species: 'Bear', habitat: 'Mountain Forest' },
  { name: 'Yogi', species: 'Bear', habitat: 'Mountain Forest' },
  { name: 'Buffy', species: 'Buffalo', habitat: 'Grasslands' },
  { name: 'Bison', species: 'Buffalo', habitat: 'Grasslands' },
  { name: 'Grazor', species: 'Buffalo', habitat: 'Grasslands' },
  { name: 'Shadow', species: 'Coyote', habitat: 'Desert' },
  { name: 'Maya', species: 'Coyote', habitat: 'Desert' },
  { name: 'Blaze', species: 'Coyote', habitat: 'Desert' },
  { name: 'Storm', species: 'Coyote', habitat: 'Desert' },
  { name: 'Luna', species: 'Coyote', habitat: 'Desert' },
  { name: 'Max', species: 'Coyote', habitat: 'Desert' },
  { name: 'Sandy', species: 'Dromedary', habitat: 'Desert' },
  { name: 'Hump', species: 'Dromedary', habitat: 'Desert' },
  { name: 'Dune', species: 'Dromedary', habitat: 'Desert' },
  { name: 'Liberty', species: 'Eagle', habitat: 'Mountains' },
  { name: 'Freedom', species: 'Eagle', habitat: 'Mountains' },
  { name: 'Tantor', species: 'Elephant', habitat: 'Grasslands' }
]
```

7. Write a MongoDB query to find animals with the species Tiger and display only their name and age, sorted by age in descending order.

A: Filter by species 'Tiger', projection of name and age, and descending sort by age

```
db.animals.find( { species: "Tiger" }, { _id: 0, name: 1, age: 1 } ).sort({
age: -1 })
```

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```
{ name: 'Shere Khan', age: 10 },
{ name: 'Shere Khan', age: 10 },
{ name: 'Tigris', age: 9 },
{ name: 'Rajah', age: 8 },
{ name: 'Rajah', age: 8 },
{ name: 'Tala', age: 7 },
{ name: 'Tala', age: 7 },
{ name: 'Kumba', age: 6 },
{ name: 'Zara', age: 5 },
{ name: 'Kiran', age: 4 }
```

8. Write a MongoDB query to insert the next animal into the animals collection:

name: "Leo", age: 7, gender: "Male", species: "Lion", habitat: "Savanna", birthplace: "Masai Mara, Kenya", parents: "Unknown", gender: "Male", Unknown", gender: "Female" caretakers: id: 501, name: "Jake Adams" }}

A: Inserting the new document using `insertOne()`:

```
db.animals.insertOne({ name: "Leo", age: 7, gender: "Male", species: "Lion",
habitat: "Savanna", birthplace: "Masai Mara, Kenya", parents: [ { name:
"Unknown", gender: "Male" }, { name:"Unknown", gender: "Female" } ], caretakers:
[{ id: 501, name: "Jake Adams" }] })
```

TEST

```
zoo> db.animals.insertOne({ name: "Leo", age: 7, gender: "Male", species: "Lion", habitat: "Savanna", birthplace: "M
asai Mara, Kenya", parents: [ { name: "Unknown", gender: "Male" }, { name:"Unknown", gender: "Female" } ], caretakers:
[{ id: 501, name: "Jake Adams" }] })
{
  acknowledged: true,
  insertedId: ObjectId('69f36c8d44d82a394d3682d1')
}
zoo> db.animals.find({ name: "Leo" })
[
  {
    _id: ObjectId('69f36c8d44d82a394d3682d1'),
    name: 'Leo',
    age: 7,
    gender: 'Male',
    species: 'Lion',
    habitat: 'Savanna',
    birthplace: 'Masai Mara, Kenya',
    parents: [
      { name: 'Unknown', gender: 'Male' },
      { name: 'Unknown', gender: 'Female' }
    ],
    caretakers: [ { id: 501, name: 'Jake Adams' } ]
  }
]
```

9. Write a MongoDB query to update the age (to 8) of the animal you just inserted using the \$set operator

A: Updating the age field using `$set` to avoid modifying other fields

```
db.animals.updateOne( { name: "Leo", species: "Lion" }, { $set: { age: 8 } })
```

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```
zoo> db.animals.updateOne( { name: "Leo", species: "Lion" }, { $set: { age: 8 } })
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
```

TEST

```
zoo> db.animals.find( { name: "Leo" })
[
  {
    _id: ObjectId('69f36c8d44d82a394d3682d1'),
    name: 'Leo',
    age: 8,
    gender: 'Male',
    species: 'Lion',
    habitat: 'Savanna',
    birthplace: 'Masai Mara, Kenya',
    parents: [
      { name: 'Unknown', gender: 'Male' },
      { name: 'Unknown', gender: 'Female' }
    ],
    caretakers: [ { id: 501, name: 'Jake Adams' } ]
  }
]
```

10. Write a MongoDB query to rename the habitat field to environment for all animals in the animals collection.

A: Using `$rename` with `updateMany()` to affect the entire collection

```
db.animals.updateMany( {}, { $rename: { "habitat": "environment" } })
```

```
{
  _id: 20,
  name: 'Yogi',
  age: 10,
  gender: 'Male',
  species: 'Bear',
  birthplace: 'Glacier National Park, USA',
  parents: [
    { name: 'Unknown', gender: 'Male' },
    { name: 'Unknown', gender: 'Female' }
  ],
  caretakers: [ { idCaretaker: 404, nameCaretaker: 'Michael Davis' } ],
  environment: 'Mountain Forest'
}
```

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11- Write a MongoDB query using the aggregation framework to find animals aged less than 10 and sort them alphabetically by their name.

A: Aggregation pipeline using to filter by age and `$match` and `$sort` to order by name:

```
db.animals.aggregate([ { $match: { age: { $lt: 10 } } }, { $sort: { name: 1 } } ]])
```

```
zoo> db.animals.updateMany( {}, { $rename: { "habitat": "environment" } })
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 71,
  modifiedCount: 71,
  upsertedCount: 0
}
```

12. Write a MongoDB query using the aggregation framework to group animals by their species and count the number of animals in each species

A: Pipeline with `$group` using `$sum` to count animals for each species

```
db.animals.aggregate([ { $group: { _id: "$species", totalAnimals: { $sum: 1 } } }, { $sort: { _id: 1 } } ]])
```

TEST

```
zoo> db.animals.aggregate([ { $group: { _id: "$species", totalAnimals: { $sum: 1 } } }, { $sort: { _id: 1 } } ]])
[
  { _id: 'Bear', totalAnimals: 5 },
  { _id: 'Buffalo', totalAnimals: 3 },
  { _id: 'Coyote', totalAnimals: 6 },
  { _id: 'Dromedary', totalAnimals: 3 },
  { _id: 'Eagle', totalAnimals: 2 },
  { _id: 'Elephant', totalAnimals: 4 },
  { _id: 'Giraffe', totalAnimals: 4 },
  { _id: 'Hippopotamus', totalAnimals: 3 },
  { _id: 'Kangaroo', totalAnimals: 3 },
  { _id: 'Lion', totalAnimals: 6 },
  { _id: 'Mandrill', totalAnimals: 7 },
  { _id: 'Panther', totalAnimals: 3 },
  { _id: 'Rattlesnake', totalAnimals: 1 },
  { _id: 'Rhinoceros', totalAnimals: 1 },
  { _id: 'Spider Monkey', totalAnimals: 3 },
  { _id: 'Tiger', totalAnimals: 10 },
  { _id: 'Wolf', totalAnimals: 7 }
]
```

13. Write a MongoDB query to create a new collection called Caretakers by extracting all caretakers from the animals collection. Ensure that caretakers do not repeat and include an array of the animals they attend to.

A: Deconstructs the caretakers array, groups by caretaker, and collects the animals they care for

```
db.animals.aggregate([ { $unwind: "$caretakers" }, { $group: { _id: "$caretakers.idCaretaker", name: { $first: "$caretakers.nameCaretaker" }, animals: { $addToSet: "$name" } } }, { $sort: { _id: 1 } }, { $out: "Caretakers" } ]])
```

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14. Write a MongoDB query to find the top 3 oldest animals.

A: Descending sort by age and a limit of 3 results

```
db.animals.find().sort({ age: -1 }).limit(3)
```

```
zoo> db.animals.find().sort({ age: -1 }).limit(3)
[
  {
    _id: 14,
    name: 'Jumbo',
    age: 30,
    gender: 'Male',
    species: 'Elephant',
    birthplace: 'Kruger National Park, South Africa',
    parents: [
      { name: 'Dumbo', gender: 'Male' },
      { name: 'Ella', gender: 'Female' }
    ],
    caretakers: [ { idCaretaker: 302, nameCaretaker: 'Emily Brown' } ],
    environment: 'Grasslands'
  },
  {
    _id: 13,
    name: 'Dumbo',
    age: 25,
    gender: 'Male',
    species: 'Elephant',
    birthplace: 'Serengeti National Park, Tanzania',
    parents: [
      { name: 'Unknown', gender: 'Male' },
      { name: 'Unknown', gender: 'Female' }
    ],
    caretakers: [ { idCaretaker: 301, nameCaretaker: 'John Smith' } ],
    environment: 'Grasslands'
  },
  {
    _id: 15,
    name: 'Ella',
    age: 20,
    gender: 'Female',
    species: 'Elephant',
    birthplace: 'Serengeti National Park, Tanzania',
    parents: [
      { name: 'Unknown', gender: 'Male' },
      { name: 'Unknown', gender: 'Female' }
    ],
    caretakers: [ { idCaretaker: 303, nameCaretaker: 'Alice Green' } ],
    environment: 'Grasslands'
  }
]
```


15. Write a MongoDB query to display animals between the 6th and 12th entries in the collection based on their insertion order.

A: Using skip() to bypass the first 5 documents and limit() to retrieve the next 7 (entries 6 to 12)

```
db.animals.find().skip(5).limit(7)
```

```
[
  {
    _id: 6,
    name: 'Zara',
    age: 5,
    gender: 'Female',
    species: 'Tiger',
    birthplace: 'Sundarbans, India',
    parents: [
      { name: 'Rajah', gender: 'Male' },
      { name: 'Tala', gender: 'Female' }
    ],
    caretakers: [ { idCaretaker: 106, n
    environment: 'Rainforest'
  },
  {
    _id: 7,
    name: 'Kiran',
    age: 4,
    gender: 'Male',
    species: 'Tiger',
    birthplace: 'Chitwan National Park',
    parents: [
      { name: 'Kumba', gender: 'Male' },
      { name: 'Tigris', gender: 'Female'
    ],
    etc...
```

etc...

16. Write a MongoDB query using the aggregation framework to create a new collection called Habitats. This collection should group animals by their environment (formerly habitat) and include the following fields.

- The environment name.
- The total number of animals in that environment.
- An array of unique species in that environment, sorted alphabetically.
- An array of unique caretakers who attend animals in that environment.

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A: Groups by environment, counts animals, retrieves unique sorted species, and unique caretakers

```
db.animals.aggregate([{$unwind:"$caretakers"},{$group:{$_id:"$environment",totalAnimals:{$sum:1},species:{$addToSet:"$species"},caretakers:{$addToSet:"$caretakers.nameCaretaker"}}},{$project:{$_id:0,environment:"$_id",totalAnimals:1,species:{$sortBy:{$input:"$species",sortBy:1}},caretakers:1}},{$out:"Habitats"}])
```

Test → Using `show collections`

```
zoo> show collections
animals
Caretakers
Habitats
zoo> |
```

Test → show all generated documents → `db.Habitats.find(). pretty()`

```
[
  {
    _id: ObjectId('69f36ffe4135e13feee97729'),
    totalAnimals: 2,
    caretakers: [ 'Emily Brown', 'John White' ],
    environment: 'Mountains',
    species: [ 'Eagle' ]
  },
  {
    _id: ObjectId('69f36ffe4135e13feee9772a'),
    totalAnimals: 10,
    caretakers: [
      'Mark Taylor',
      'Emily Brown',
      'John Smith',
      'Alice Green',
      'Michael Davis',
      'Sophia Green'
    ],
    environment: 'Savanna',
    species: [ 'Giraffe', 'Lion' ]
  },
  {
    _id: ObjectId('69f36ffe4135e13feee9772b'),
    totalAnimals: 23,
    caretakers: [
      'Alice Green',
      'Nathan Green',
      etc.
    ]
  }
]
```